The Analytic S-matrix Program with a Geometric Viewpoint
Part B-2

4 CV of the researcher

PERSONAL DETAILS:

First Name:	Chia-Kai	Surname:	Kuo
Date of birth:	November 14rd 1995	Citizenship:	Taiwan
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ACADEMIC POSITION:

- **Max Planck Institute for Physics**, Munich, Germany
  Postdoctoral Researcher 01/06/2024 – Present

EDUCATION:

- **National Taiwan University**, Taipei, Taiwan
  Ph.D. in Theoretical Physics, 01/02/2021 – 30/05/2024
  Supervisor: Song He, Yu-Tin Huang
  Thesis: Loop Amplitudes of ABJM and Its Positive Geometry.
  My Ph.D. research focused on the on-shell approach to ABJM theory, in particular the formulation of scattering amplitudes as differential forms associated with novel positive geometries.
  M.Sc. in Physics. 01/09/2018 – 31/08/2020
  B.Sc. in Physics. 01/09/2014 – 31/08/2018

PUBLICATIONS:

I have authored **10** original research articles in peer-reviewed journals and **1** preprint articles, with **178** citations and a h-index **7**. For up-to-date values, see https://inspirehep.net/authors/1945409

Refereed Publications

[1] C.-K. Kuo and Q. Yang, "Notes on off-shell conformal integrals and correlation functions at five points" *Phys.Rev.D* **113** (2026) 5, [arXiv:2512.21947].

Description: We construct pure two-loop five-point off-shell conformal integral bases and use them to compute five-point half-BPS correlators at symbol level, including both maximal and non-maximal sectors.

[2] Y.-T. Huang, C.-K. Kuo and C. Zhang, "BDS ansatz in ABJM via scaffolding triangulations" *JHEP* **02**, (2026) 022, [arXiv:2510.20663].

Description: We formulate the ABJM two-loop BDS ansatz using scaffolding triangulations, proving elliptic-cut cancellation and triangulation independence at arbitrary multiplicity.

[3] S. He, Y.-T. Huang, and C.-K. Kuo, "Leading singularities and chambers of Correlahedron" *JHEP* **03**, (2026) 071, [arXiv:2505.09808].

Description: We analyse leading singularities and chamber structures of the Correlahedron, showing that four-point correlator integrands admit a chamber-based geometric organisation beyond three loops.

[4] S. He, Y.-T. Huang, and C.-K. Kuo, “All-loop geometry for four-point correlation functions”, *Phys.Rev.D* **110** (2024) 8, L081701 [arXiv:2405.20292].

Description: We propose an all-loop positive geometry for four-point correlators in planar $\mathcal{N} = 4$ SYM and verify that its canonical forms reproduce known correlator integrands up to three loops.

[5] S. He, Y.-T. Huang, and C.-K. Kuo, “The ABJM Amplituhedron”, *JHEP* **09**, (2023) 165, [arXiv:2306.00951].

Description: We construct the ABJM amplituhedron as an all-loop, all-multiplicity positive geometry and obtain explicit one-loop and two-loop integrands from its canonical forms.

[6] S. He, C.-K. Kuo, Z. Li and Y.-Q. Zhang, “Emergent unitarity, all-loop cuts and integrations from the ABJM amplituhedron” *JHEP* **07**, (2023) 212, [arXiv:2303.03035].

Description: We prove all-loop unitarity properties of the four-point ABJM amplituhedron and integrate its negative geometries up to two loops to extract infrared-finite data.

[7] S. He, C.-K. Kuo, and Z. Li, “The two-loop eight-point amplitude in ABJM theory” *JHEP* **02**, (2023) 065, [arXiv:2211.01792].

Description: We compute the two-loop eight-point ABJM amplitude, fixing the integrand from dual conformal symmetry, maximal cuts and soft-collinear constraints, and analyse its integrated finite part.

[8] S. He, C.-K. Kuo, Z. Li and Y.-Q. Zhang, “All-Loop Four-Point Aharony-Bergman-Jafferis-Maldacena Amplitudes from Dimensional Reduction of the Amplituhedron”, *Phys. Rev. Lett* **129** (2022) 221604, [arXiv:2204.08297].

Description: We define a reduced amplituhedron for four-point ABJM amplitudes by dimensional reduction of the $\mathcal{N} = 4$ SYM amplituhedron and obtain explicit integrands up to five loops.

[9] S. He, C.-K. Kuo and Y.-Q. Zhang, “The momentum amplituhedron of SYM and ABJM from twistor-string maps”, *JHEP* **02**, (2022) 148, [arXiv:2111.02576].

Description: We relate twistor-string maps to momentum amplituhedra in $\mathcal{N} = 4$ SYM and ABJM, showing how tree amplitudes arise as canonical forms of positive geometries.

[10] Y.-T. Huang, C.-K. Kuo and C. Wen, “Dualities for Ising networks”, *Phys. Rev. Lett* **121** (2018) 251604, [arXiv:1809.01231].

Description: We establish a correspondence between planar Ising networks and cells of the positive orthogonal Grassmannian, leading to recursive methods for computing network correlators.

Preprint Publications

[11] Y.-T. Huang, C.-K. Kuo, Y. Liu and J. Mei, “Beyond Discontinuities: Cosmological WFCs from the Supersymmetric Orthogonal Grassmannian”, [arXiv:2604.08512].

Description: We extend the orthogonal-Grassmannian construction from discontinuities to full supersymmetric cosmological wavefunction coefficients, using $\mathcal{N} = 2$ supersymmetry to solve the inhomogeneous Ward identities.

AWARDS AND HONORS:

- NTU-CTP Best Student Paper Award 2023
  Awarded by the Center for Theoretical Physics at National Taiwan University for an outstanding student research paper in theoretical physics.

- NCTS Student Outstanding Paper Award 2023
  Awarded by the National Center for Theoretical Sciences for the quality and originality of student-led research in theoretical physics.

INVITED TALKS:

16 invited talks at International conferences and leading research institutions:

- **Invited Talk**: "Chamber Dissection of the Correlahedron in Four-Point Correlators",
  New Ways to Higher Points,
  Humboldt University of Berlin, Germany 01/10/2025
- **Invited Talk**: "BDS Local Integrand in ABJM Theory",
  42nd International Conference on High Energy Physics,
  IUPAP C11 and Czech academic institutions, Czech Republic 19/07/2024
- **Invited Talk**: "Geometry of the four-point correlators",
  Kick-off workshop on Positive Geometry in Particle Physics and Cosmology,
  Max Planck Institute for Mathematics, Germany 16/02/2024
- **Invited Talk**: "The ABJM Amplituhedron",
  Amplitudes 2023 Conference,
  CERN, Switzerland 07/08/2023
- **Invited Talk**: "All-Loop Four-Point ABJM Amplitudes from Dimensional Reduction of the Amplituhedron",
  2022 NTU-Kyoto high energy physics workshop/Kawai Fest,
  National Taiwan University, Taiwan 18/12/2022
- **Seminar**: "On-shell method on ABJM two-loop amplitudes",
  NTU-NCTS String Seminar,
  National Taiwan University, Taiwan 25/11/2022

TEACHING AND SUPERVISION EXPERIENCE:

- Teaching Assistant for "Electricity and Magnetism", "Quantum Mechanics", and "Applied Mathematics I" at the Department of Physics, National Taiwan University.

COMPUTER SKILLS:

Languages:	LaTeX
Software:	Mathematica, Maple